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ORIGINAL ARTICLE

A revision of seismicity models based on complex systems and earthquake networks

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Abstract Statistical laws such as the Omori and Gutenberg-Richter laws have become one of the most important tools for seismic hazard. The introduction of earthquake networks by Abe and Suzuki has opened a new research field in seismology, which gives an option to view the seismicity as a complex phenomena in an approachable way and may help to improve seismic hazard assesment. This paper intends to make a revision of the earthquake network models proposed in the literature, and to look for open questions by the date which may be of interest for future research.

Keywords Earthquake networks · Seismicity

1 Introduction

Through the statistical analysis of seismicity, two of the most important laws of seismology have been formulated: the Omori law (Utsu et al. 1995) which describes the frequency of occurrence of aftershocks after a great earthquake, and the Gutenberg-Richter law (Gutenberg and Richter 1944) which describes the general frequency of occurrence of events with different magnitudes. These two laws involve some behaviors that follow power laws, which are common in complex systems. As a result, different investigations of the seismic phenomenon have been carried out expressing it as a complex system, such as the formulation of models of the earthquake nucleation (Olami et al. 1992; Bak and Tang 1989) (which is, how an earthquake is generated) or seismicity in general as complex networks (Abe and Suzuki 2004b; Baiesi and Paczuski 2004; He et al. 2014; Rezaei et al. 2018; Telesca and Lovallo 2012). In particular, with the presentation of a sequence of seismic events as a complex network, an attempt has been made to understand the structure of these earthquake networks, and in these formulations, it has been specified that seismicity can be seen as a small-world network (Abe and Suzuki 2004b; Chorozoglou et al. 2018; León et al. 2018) and as a Scale-Free network (Abe and Suzuki 2004a; Baiesi and Paczuski 2004). For each of these cases, it has been reviewed the distribution of the degrees, the links, and the effect that large events can generate on the structure of the networks. In this regard, the dynamic analysis of earthquake networks has allowed see some changes in network parameters before and after strong earthquakes (Abe and Suzuki 2007, 2012; Chorozoglou et al. 2018; Pastén et al. 2016; Pastén et al. 2018).

Complex networks are a topic of research of great interest. Much of the generalities of complex networks have been commented by Boccaletti et al. (2006),

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where characteristics used to analyze networks such as clustering, degree distribution, connectivity, centrality, and path lengths are described. Some of the research lines in complex networks that may be of interest for seismology have to do with predictions; hence, it is relevant the prediction of links in complex networks using probabilistic and similarity methods (Lü and Zhou 2011) which may help in prediction tasks in an analysis of seismicity.

The purpose of this document is to collect a good part of the efforts that have been made in the analysis of earthquake networks to know what questions remain to be resolved in this area of research. The paper will discuss some theoretical foundations that support seismicity analysis as a complex phenomenon and complex network analysis. Then, we try to show the methodologies used to build earthquake networks, the most relevant characteristics used in the analysis of these networks and the information that these measured parameters of the networks provide about seismicity. Finally, we gather some conclusions and ask some questions that can serve as motivation for future works.

2 Complexity in seismicity and complex networks

The work of Omori (Utsu et al. 1995) and Gutenberg and Richter (1944) on the behavior of seismicity resulted in two of the most important laws of seismology, which describe the frequency of occurrence of aftershocks after an earthquake, as well as the frequency of occurrence of events with different magnitudes. Since in both laws there is a power law behavior, the complexity of the seismicity phenomenon is evidenced. In the work of Bak and Tang (1989), seismicity is modeled as a phenomenon with critical self-organization using blocks linked by springs, and the power laws of Gutenberg-Richter and Omori are related to behaviors of critical self-organization that emerge from such system. This work defines an initial framework to study seismicity as a complex phenomenon, which continues with the work of Ito (1992), discussing fractality and multifractality in seismicity. In the preface of "Introduction to Nonextensive Statistical Mechanics" by Tsallis (2009) it is said that "*One of the main – necessary but by no means sufficient – features of complexity has to do with the fact that both very ordered and very disordered systems are, in the sense of plectics, considered to be simple, not complex*" (*Plectics* has to be understood as the science that studies complexity). In this work are described some of the analytical tools used in complex systems, and it is also explained that the entropy of such systems differs from the entropy of systems of the random type such as fluid or chaotic in the sense that the total entropy of the system is not the sum of the entropy of its parts, which requires a *non-extensive statistical mechanics*. This type of tools, applied to seismology, derives in a branch that can be described as non-extensive statistical seismology, collected by Vallianatos et al. (2018), where many of the phenomena that make seismicity seen as a complex phenomenon are described, such as scaling relationships, spatial and temporal correlations, and critical phenomena; as well as the way to apply non-extensive statistical mechanics in seismology.

Boccaletti et al. presented a state of the art (Boccaletti et al. 2006) about calculations made in complex networks; numerous parameters and measures related to complex networks are described, as well as characteristics of real networks. Many of the characteristics inherent to complex systems are found in complex networks, such as self-similarity, fractality, and scalability. There are different types of complex networks. The most important ones can be classified as *small-world*, *scale-free*, and *random* (random networks are referenced in the literature as *Erdös-Renyi* networks). Completely random networks began to be studied by Erdos and Renyi (1960), which became a first effort to lay the foundations with which the networks were studied later. small-world networks were coined by Watts and Strogatz (1998), and are presented as a type of networks that lie between regularly connected networks and networks connected completely randomly. Scale-free networks are a type of networks that demonstrate a distribution that follows a power law in the number of connections that nodes have (known as the degree of a node), and have the property of having a few nodes with many connections and many nodes with few (Barabási 1999). Real-world networks are normally compared to and studied from the perspective one of these three type of networks. Table 1 is taken from Chorozoglou et al. (2019), and some of the parameters of the networks that are most used for analysis are described.

Table 1 Seven network measures in complex networks. Taken from Chorozoglou et al. (2019)

Network parameter	Description
Degree centrality	Average over all nodes of either the number or the strength of connections.
Clustering coefficient	Average over all nodes of the fraction of connections between the nodes within their neighborhood, divided by the number of all possible connections.
Characteristic path length	Average of the shortest path lengths in the network, computed over all pairs of nodes.
Global efficiency	Average inverse shortest path length in the network, computed over all pairs of nodes.
Betweeness centrality	Average over all nodes of the node betweenness centrality, which is the fraction of all shortest paths in the network that contain the node, divided by the number of all possible paths.
Modularity	Shows the strength of a network that is divided into components.
Eccentricity	Average over all nodes of the maximal shortest path length between a node and any other node.

Some complementary parameters have been proposed to analyze networks, such as small-worldness (which, for a network i is defined as $S = \frac{\frac{C}{C_r}}{\frac{L}{L_r}}$, and it is calculated comparing the clustering and the path lengths in the network with the same parameters of an Erdös-Renyi random network) (Humphries and Gurney 2008), or Renyi's entropy (Duan et al. 2019). Wiedermann et al. (2017) show a way to characterize different types of complex networks using the entropy calculated from the distribution of node degrees, and statistical complexity calculated from the adjacency matrices of the network, as a product of entropy and divergence of the network from a state of equilibrium, which for the purposes of complex networks is chosen as a similar random network. By plotting different networks on a plane whose coordinates are these two parameters (entropy and complexity), different types of networks can be located in different places of the entropy-complexity plane; delivering a tool that allows to compare different types of networks. Other types of entropies can be reviewed, as entropy can be estimated from the distribution of any measure from the nodes, such as degree, centrality, or average path length. While the calculation of entropy is valid to check the amount of information in a system, using Shannon's entropy may not be the best option in earthquake networks, since as commented by Vallianatos et al. (2018) seismicity has properties which can be better described using non-extensive statistical mechanics, so the use of Tsallis entropy could be more appropriate to describe such systems.

3 Earthquake networks

A sequence of earthquakes is a series of events that occur in space and time. Gao et al. (2016) present a state of art regarding different methods with which complex networks can be constructed from time series, such as visibility graphs (several types of these graphs are listed), recurrence networks, and networks based on transitions between other methods. It is commented that this type of analysis could open a place for cross-fertilization between methods used to analyze time series and the theory of complex networks. It is interesting about this article regarding the research proposed in earthquake networks that as earthquakes follow a temporal sequence, and so a seismicity catalog can be seen as a time series of events, they can be expressed as complex networks in ways than have not yet been contemplated in the literature. The joint use of complex networks and time series is also evident in the work by Pastén et al. (2018), which shows a procedure to transform a complex network of earthquakes into a time series through a non-random walk, following the sequence of occurrence of earthquakes. The time series constructed from this connectivity shows changes in multifractality due to characteristics in the studied areas, and that in general multifractality decreases when seismic activity increases. This work shows that there are other parameters that can be used to discover not so obvious patterns of seismicity, since in one of the areas studied, the increase in multifractality after large earthquakes was very different than in another area because one of the earthquakes had many more aftershocks than the other. Reviewing complex

Table 2 Rules used to define nodes and links in earthquake networks

Definition of nodes	Definitions of links
Region divided in cells, and a cell is chosen as a node if an earthquake occurs into it.	Cells are linked following the temporal sequence of earthquakes Abe et al. (2004a, b, 2011), Chorozoglou et al. (2017), Leon et al. (2018).
	Cells are linked if a visibility condition between nodes is met, based on the magnitudes and the time between events (Rezaei et al. 2018).
	Cells are linked if it is met that an earthquake in the following cell is inside a space-temporal influence domain for the previous event in the first cell (He et al. 2014).
Events as nodes	Events are linked if a visibility criteria is met based on the magnitude and the time between events (visibility graph) (Telesca and Lovallo 2012).
	Events are linked if they are inside the space-temporal influence domain (Xu et al. 2018).
	Events are linked using a correlation factor from magnitudes, occurrence times and distances (Baiesi and Paczuski 2005; Suteanu 2015).

networks as time series seems to generate very interesting information about the activity and deserves to be taken into account as an analysis tool.

Abe and Suzuki (2004a) were pioneers in the analysis of seismicity using complex networks. The constructed earthquake networks contain various properties that can be directly related to parameters estimated from traditional seismicity analyzes, showing that the analysis of earthquake networks can be a tool that allows to analyze the complexity of the seismicity. From that work, different ways of building earthquake networks have been developed, which can be differentiated in the way in which nodes are defined. On the one hand, there are earthquake networks that are constructed by dividing a region into cells and a cell is selected as a node as long as a seismic event occurs in it. On the other hand, there are earthquake networks in which the earthquakes are the nodes. Table 2 presents different methods of creating earthquake networks, indicating how the nodes are defined, the rules governing the links between nodes, and it is indicated in which works such networks have been used.

3.1 Cells as nodes

Abe and Suzuki (2004b) describe a way to create an earthquake network from a seismicity catalog, performing a division of the studied region in cubic cells which are chosen as nodes of the network if earthquakes occur within the cells, and are connected following the temporal sequence of the earthquakes. Figure 1 is taken from León et al. (2018), and it is shown how these networks are constructed.

The observation of the earthquake networks reported by Abe and Suzuki (2004a) for seismicity in California and in Japan gave as a result that these networks have a degree distribution which follows a power law, so it was seen that these are scale-free networks: result that somehow agrees with the Omori and Gutenberg-Richter laws, to the extent that large earthquakes that have many aftershocks function as hubs of these networks. Earthquake networks can be built with or without the directivity of the connections; however, since in the case of directed networks the nodes have the same input and output degree, the networks can be analyzed as non-directed. Also, links of a node with itself (loops) as well as multiple or a single link between two nodes may or may not be taken into account to analyze the small-world nature of the earthquake network (Abe and Suzuki 2004b; León et al. 2018). Abe et al. (2011) analyze earthquake networks from both perspectives; and it can be seen that the clustering in these networks is much greater and that the paths are shorter than in random networks. In this work, an analysis of the cell size variation is made, which shows that clustering and connectivity remain constant ($C \sim 0.85$ and $\gamma \sim 1.0$) after exceeding a given size, which is a consequence of the cells grouping more events (number of nodes decrease), but the nodes have greater connectivity (the links on certain nodes increase). In this work, it can also be seen that there are global seismic behaviors which

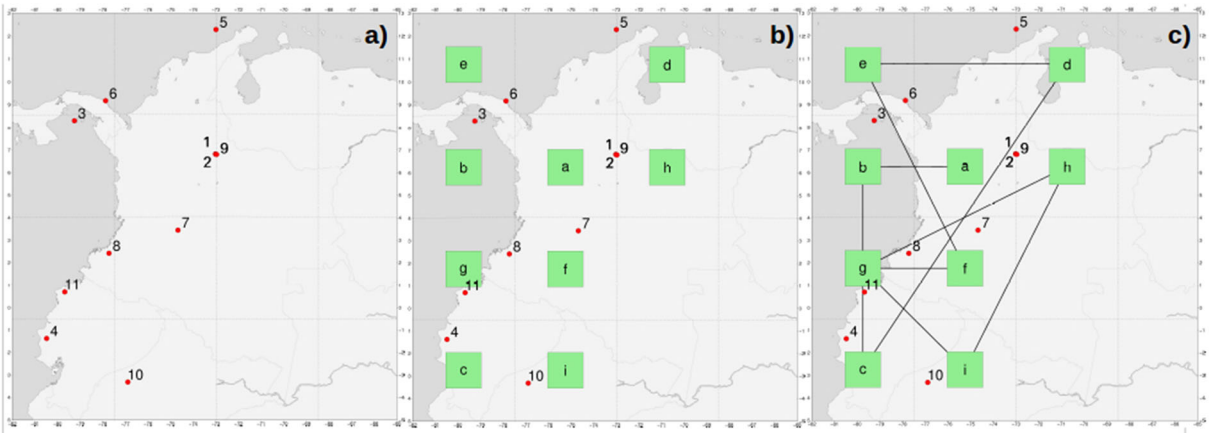

Fig. 1 Diagram of the earthquake network produced by the spatiotemporal sequence of earthquakes. **a** The sequence of events (numbered dots) is mapped on a grid. The number near the events follows the temporal order from oldest to newest. **b** A cell from the grid is selected as a node if there are events mapped inside the cell. The letter inside the node (squares) identifies the node. **c** Nodes are connected following the temporal sequence of events, linking two nodes with successive events. Taken from León et al. (2018)

can be evaluated from the point of view of complex networks.

3.2 Earthquakes as nodes

Another way to build the earthquake networks is to link the earthquakes between them based on different rules. There are different criteria that have been used to link earthquakes, ranging from visibility graphs, correlation of events or belonging to the same domain of space-time influence. Telesca and Lovallo (2012) propose to build earthquake networks based on visibility graphs based on the magnitude of the events and the temporal distance between them. Figure 2, inspired from that same article, shows how these types of graphs are constructed.

Visibility graphs were introduced by Lacasa et al. (2008), as a simple computational tool to study series from the perspective of graphs, connecting different nodes from some measure of amplitude. This type of graph retains many characteristics of the time series they represent, so Telesca and Lovallo in their work introduce this technique to review the series of earthquake magnitudes, showing that the distribution of degrees follows a power law.

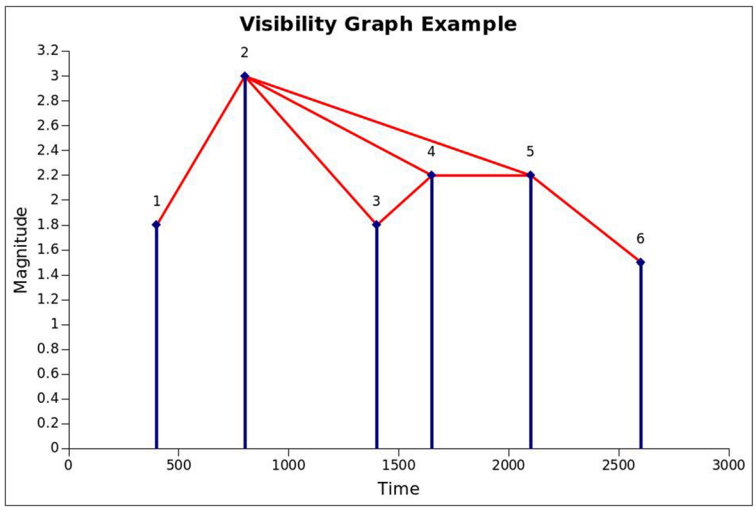

Fig. 2 Earthquake network constructed from a visibility graph. Earthquakes are mapped as dots in a magnitude vs. time plane; and links are made so they dont intersect any vertical line from the dots and the time axis. Inspired from Telesca and Lovallo (2012)

Baiesi and Paczuski (2005) describe another way to create a directed earthquake network from a certain earthquake to subsequent earthquakes using an event correlation metric.

Correlation between two events is defined as $c_{ij} = 1/n_{ij}$, where $n_{ij} \equiv K t l^{d_f} 10^{-bm_i} \Delta m$. The correlation metric takes into account the difference of the magnitudes Δm, the distance l, and the time t that occurs between the events, as well as parameters derived from the analysis of the Gutenberg-Richter curve in the study region (parameter b), the fractal dimension of seismicity d_f, and the calculation of a constant K which is calculated for a region, time interval, and underlying seismicity. Paradoxically, this constant depends on many factors, so it would be interesting to make the correlation metric more robust by directly taking the values on which it depends instead of estimating the entire catalog, since it is known the number of earthquakes, the b parameter and the magnitude differences are highly dependent on the occurrence of large events. In the end, the paper shows that the networks built with this method are of the scale-free type, and concludes that the correlation threshold seems a bit arbitrary and generates different networks that, in any case, maintain in their structure a constant exponential parameter that scales the magnitude of an earthquake with the distance of each replica. It also concludes that this approach serves as a sign that some universal properties of networks may exist independent of the type of metric used.

Suteanu (2015) propose a way to create networks of earthquakes similar to that proposed by Baiesi and Paczuski (2005), for volcanic events in Hawaii; however, the links have a weight which is a combination of spatial, temporal, and magnitude variables. In the formulas used to estimate the weights, constants are used which, for the purposes of the study, are tested with different values. These two studies motivate, in some way, the search for parameters and metrics of seismic networks that can characterize seismicity.

Xu et al. (2018) propose another way of building earthquake networks similar to that proposed by Baiesi and Paczuski. In these networks, a correlation threshold is not used as such, but it is stated that each earthquake has a threshold of influence in time and distance that depends on the magnitude, whereby links are generated between events that are within the domain of that space-time influence. By using this method, the division by node communities was explored, concluding that using the communities can slightly improve the prediction of events (although the accuracy of the predictions is still below 0.5).

3.3 Seismicity information from earthquake networks

Measurements from the earthquake networks have a different meaning, depending on the type of earthquake network constructed. For instance, the degree in a *cells as nodes* network implies how many times an earthquake can occur in the sector which corresponds to that node after an earthquake has occurred on other sectors, so a high degree for a node implies many earthquakes in a sector. For the *earthquakes as nodes* networks, degree implies how many earthquakes are inside its influence radius (that is, how many earthquakes can be related as aftershocks or foreshocks), so a high degree for a node implies an important earthquake in terms of the quantity of foreshocks and aftershocks. The meaning of the other parameters merits a more detailed analysis respect to the implications in the seismicity dynamics.

From the analysis of the parameters found in earthquake networks, some conclusions found in the literature can be summarized on Table 3.

Other conclusions that emerge from other works can be listed below. Chorozoglou et al. (2017) made a review of the evolution of the earthquake networks created using the Abe and Suzuki method before and after large earthquakes, analyzing the cases with and without weights in the links, and using 30-day windows which move in periods comprising 300 days before and 150 days after a large earthquake. The clustering coefficient, eccentricity, small-worldness, path length, and efficiency of the networks are reviewed (efficiency in a network can be seen as the inverse of the path length, and indicates how effective is the transmission of data by the network), and the calculation of a statistical factor p in which the earthquake network is compared with similar random networks in the degree distribution (for the case without weights) and in the connectivity (for the case with weights). The conclusions show that in the case of links with weights, there is a behavior that seems to announce in a short time the appearance of a large earthquake, and an increase in centrality is appreciated before a large event. In this work can be seen that earthquake networks could be applied as short-term prediction tools (a few months) before a large earthquake, although it

Table 3 Earthquake network parameters

Network parameter	Seismicity information
Community division	Earthquake prediction by the network can be improved using a community division (Xu et al. 2018).
Quantity of links	It is shown that the Gutenberg-Richter data can be estimated using the Abe-Suzuki model and the Omori law using the Telesca-Lovallo model (Rezaei et al. 2017)
Connectivity distribution	It follows a power law, which agrees with the Gutenberg-Richter law; showing the presence of giant components (earthquakes that generate aftershocks), and a behavior of *preferential attachment* (Abe and Suzuki 2006).
Multifractality	Multifractality is evaluated in a temporal series created from an earthquake network, it is shown a chance by the presence of strong earthquakes (Pastén et al. 2018).
Small-worldness and clustering	Changes in small-worldness and clustering can be of use to indicate a possible seismic risk in a short term (Chorozoglou et al. 2017).

is noted that it is important to continue investigating to corroborate this statement.

Deyasi et al. (2017) made another review of earthquake networks created with the Abe and Suzuki method, for different regions in the world that are located between different tectonic plates. A measurement of the similarity of the networks from the Jensen-Shannon divergence is used, and it was observed that seismic regions near the same tectonic plate can generate similar networks. The paper raises the difficulty of measuring the distance between two earthquake networks built with different amounts of seismic events and generating networks with different numbers of nodes. This difficulty was addressed by calculating the spectrum of adjacency matrices, and although it is not mentioned, it is shown in the article that the structure of the networks have some characteristics which are conditioned by the tectonics of the region and that this conditioning covers the regions that are close to such tectonic structures.

Ohsawa (2018) presents a method to detect foreshocks of large events using a Regional Entropy of Seismic Information index. Entropy, as in almost all work done in complex systems, is evaluated from the point of view of the amount of information in a system. The method is applied in seismic clusters, which are created by dividing the study region into a mesh and the cumulus sizes in each mesh cell are estimated. It is observed that some gaps appear in the seismicity before the appearance of a large earthquake, which has been seen to correspond to changes in the behavior of this entropy index; and thus the index can detect a precursor activity up to a year before a large earthquake. Although in this work a seismic network is not created, the idea of using the changes in the entropy of the system, that is, the changes in the information of the seismic system, shows promising results for an earthquake prediction system, which could well be used in an earthquake network where the calculation of entropy is easier.

About what is the right size to build a cells-as-nodes earthquake network, some works have been found. Lotfi and Darooneh (2012) revise a resolution parameter which is inversely proportional to the size of the cells, and it is concluded that when the resolution increases, the characteristics of the networks become very dependent on each other. Also, Abe and Suzuki (2009) reviewed the cell size, and it is found that increasing the cell size there is a particular size in which the clustering ceases to have considerable variations (about 0.85 for several regions that were studied).

4 Conclusions and discussion

From all these elements, the possibility of building a seismicity model from the perspective of complex networks is raised. The following are questions that remain open:

What is the best cell size to create an earthquake network of the type cell as a node? From the representation with all the links (both with loops and multiple links between nodes) in the seismic networks of the cell type as node, how much can be predicted about the

seismological activity? And how much from the simplified representation that evaluates the small-world characteristics of earthquake networks? Entropy, complexity, and divergence as evaluated by Wiedermann et al. have not been evaluated in detail in earthquake networks. How much information do these parameters provide about seismological activity? In seismology, the analysis of the dynamics of mechanical stresses (internal forces within the earth) as a result of seismological activity is of great interest. How can a model that relates earthquake networks and the mechanical stress release due of the occurrence of earthquakes be created?

For the authors, what has to do with a model that relates stress release with earthquake networks can be very complicated and it would be very interesting to think of some first steps to outline a model of this type. However, based on what is already known about earthquake networks, another research question which can be raised that brings together several of these issues in this way:

How can a dynamical model of seismicity can be developed from the representation of an earthquake network?

The research question has several questions implicit that should be answered. The first has to do with what are the best ways to create earthquake networks and what information each one delivers. The problem could be limited by reviewing only the representation of networks of the cells as nodes type, and the problem of cell size should be reviewed. The type of networks constructed has to be reviewed; hence, the statistical divergence used by Wiedermann et al. can be used as a similitude measurement respect the three main types of networks. Also, it has to be answered which are the more appropriated measures from the networks to create a dynamical model of seismicity. There are many that have been used in previous works, but other measures can be used such as the Renyi's entropy used by Duan et al. (2019), and the Tsallis-Shannon entropy used by Yulmetyev et al. (2004): or based on the distribution of other measures such as centrality, clustering, or path lengths which can involve different scenarios (in this respect it can be remembered that the entropy seen from Tsallis' vision generated the branch of non-extensive statistical mechanics Tsallis 2009). The proposed model should somehow describe some characteristics of the dynamics of seismicity, so the measurement of time in this sense becomes important in the formulation of the model. Although it cannot be specified from now on whether specific seismic events can be predicted with the proposed model, it is expected that with this model some kind of behavior can be foreseen; and this is already a sufficient motivation to say that this research is relevant: every step that is taken in the way of understanding the seismic phenomenon and that can announce a seismicity behavior will be very important for science and in particular for the nearby population to the region studied. It is expected that the proposed model can be used in different regions, and just as with the Gutenberg-Richter and Omori laws, some parameters may change depending on the region or the type of events that are presented, so it is expected that the model can characterize a seismic region and allow comparison with the model applied to the seismicity of other regions. In this regard, the question also arises of which regions should be analyzed to raise the seismicity model.

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